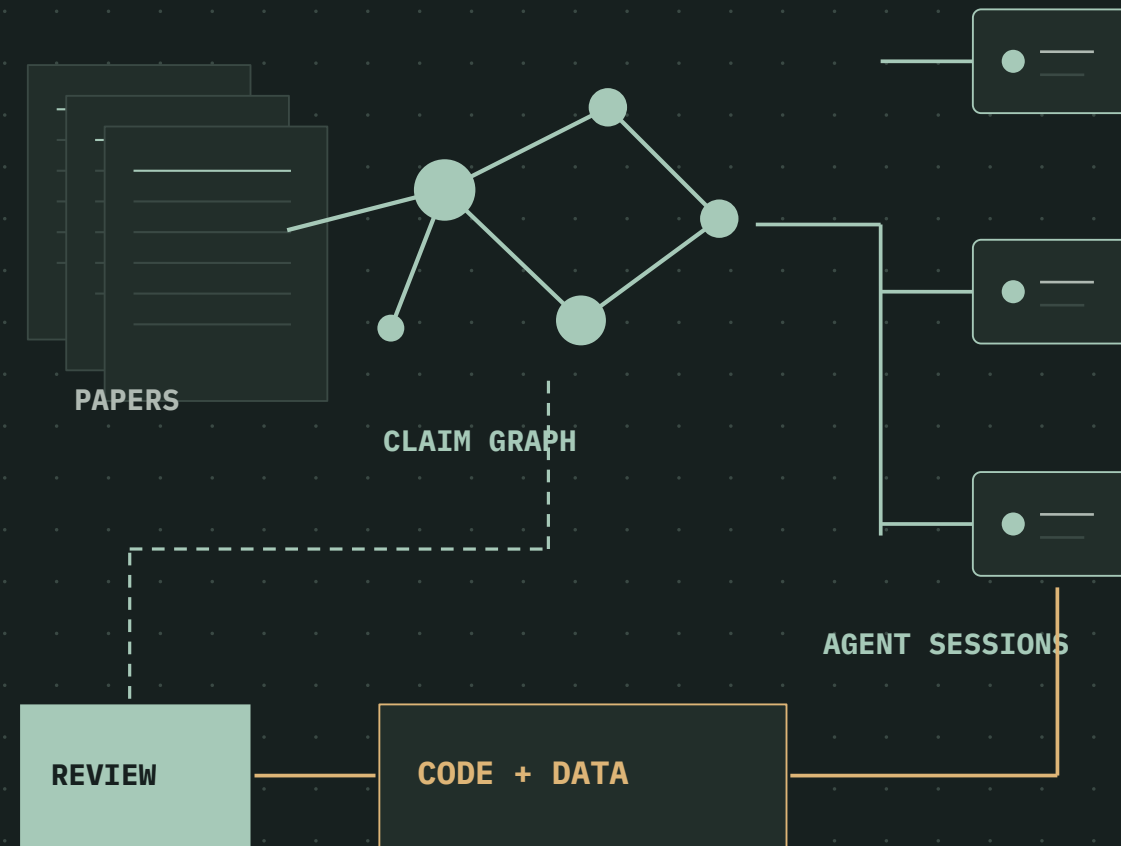


Autonomous research for biological foresight

Biological breakthroughs can create
unilateral power before defenses adapt.

We connect the literature to executable
experiments and inspectable evidence.

SOURCE → STATE → EXPERIMENT → REVIEW



Persistent evidence + bounded compute

AI is accelerating biology. Defense needs a head start.

CAPABILITY

Up to 60%

above PhD baselines on selected
biology / chemistry questions

AISI · open-ended QA benchmarks

SAFEGUARDS

198,000

adversarial attempts still exposed
one high-risk vulnerability

Anthropic · Constitutional Classifiers++

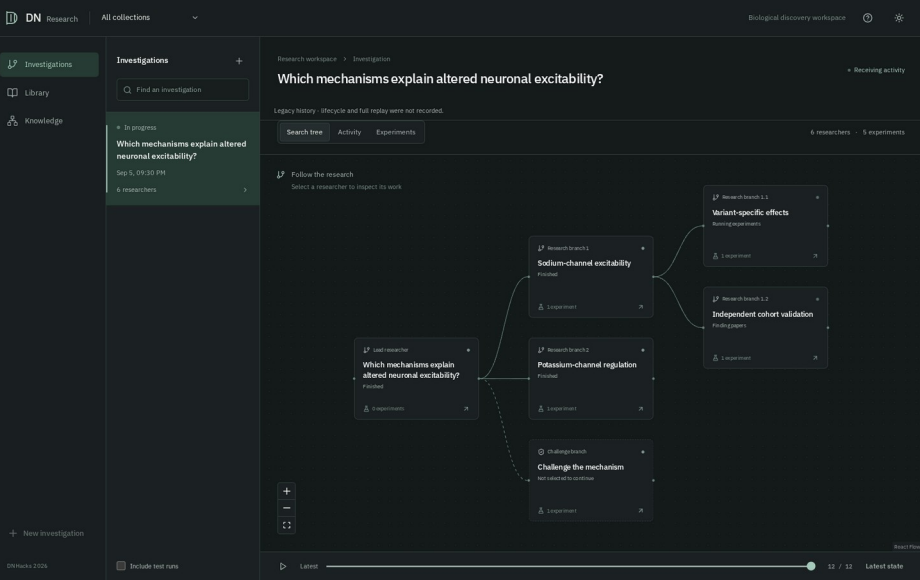
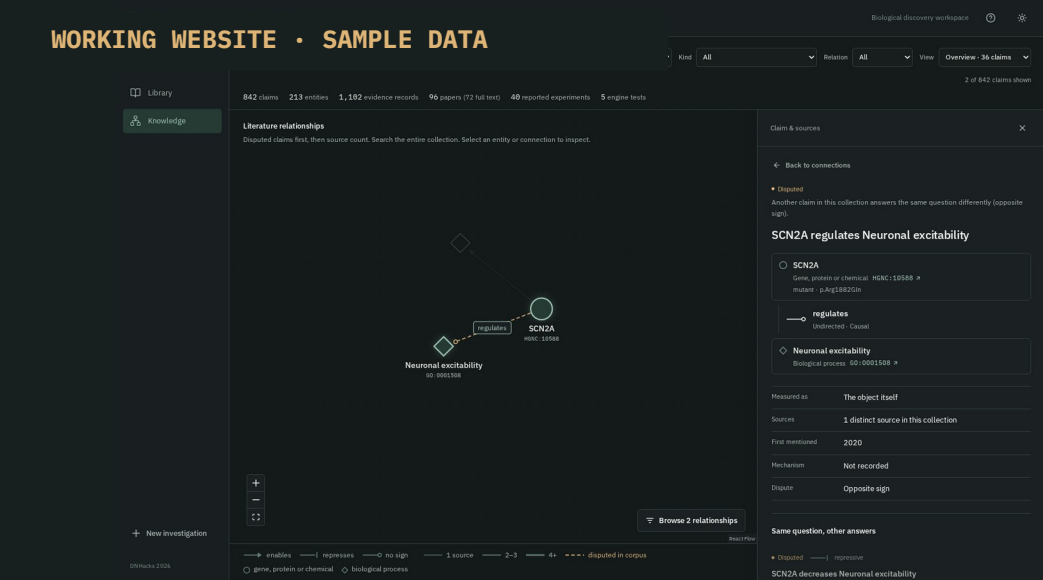
“The United States tends to play catch-up after critical technologies have already become mainstream.”

NSCEB
2025 · §1.3

A claim graph drives a persistent tree of experiments.

Ingest → grounded graph → Recursive agent sessions → Acceptance + review

HGNC / ChEBI / MONDO → canonical IDs One action → execution → observation Typed checks / artifacts / human decision



Knowledge: canonical identity, context and source inspection Investigation: branches, activity and experiment replay

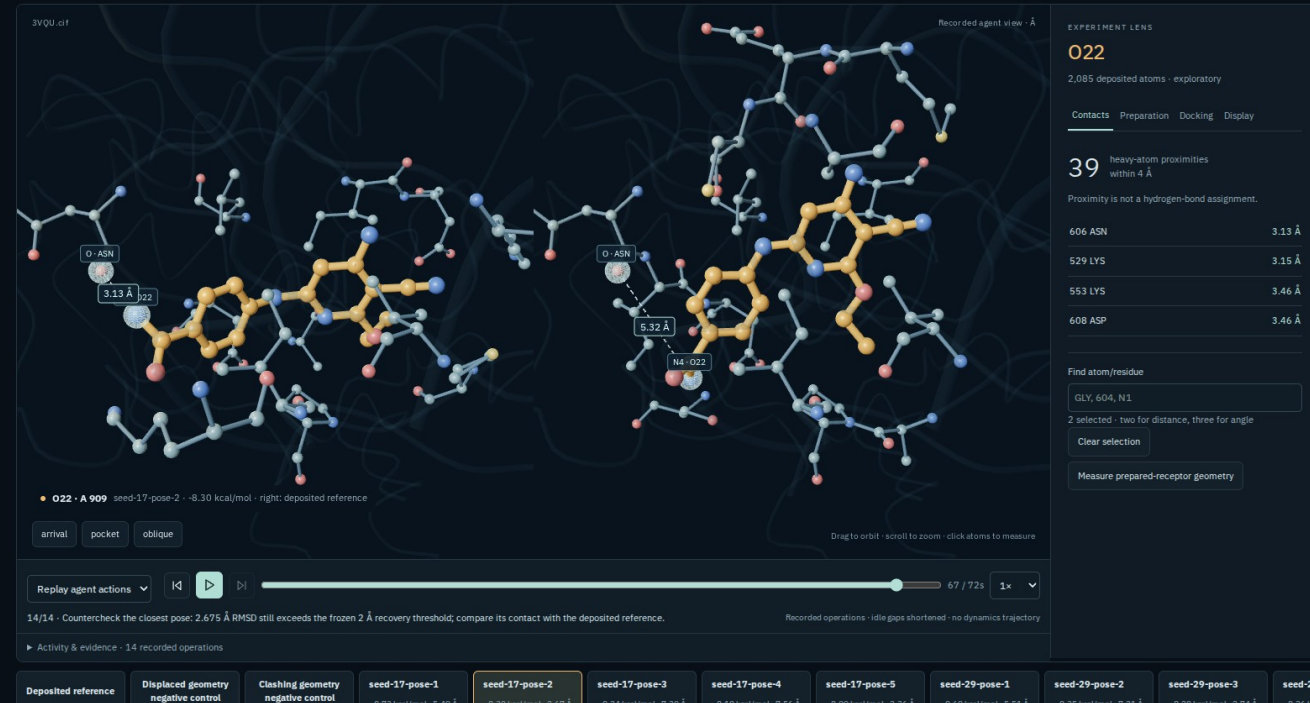
18 actions / checkpoint • 2,880 actions / shared tree • atomic grants survive restart

Images become evidence when checked against coordinates.

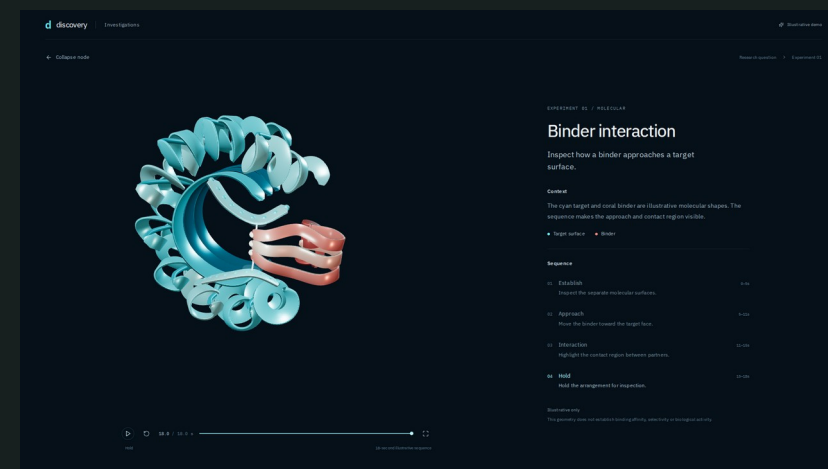
RECORDED TOOL RUN

3VQU / Binding pocket

O22 - inspect the structure, compare poses, test a contact.



Docking → image review → coordinate countercheck



OPEN VIDEO

23 seconds

Auto-plays in Slide Show · MP4 in delivery folder

Binder cinematic: illustrative geometry

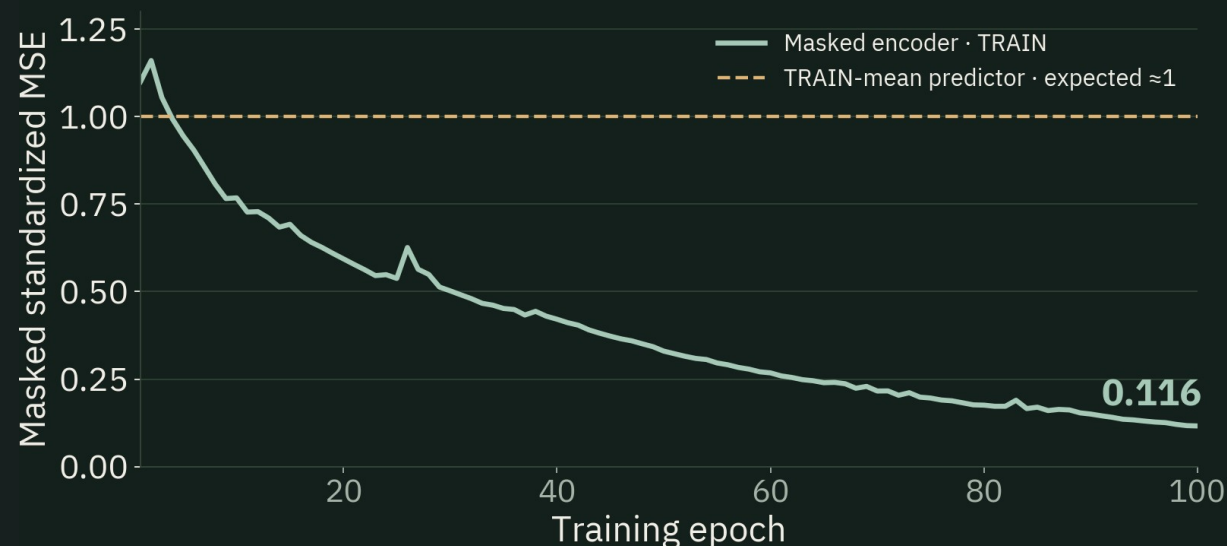
MEASURED CONTROL

2.675 Å RMSD / required ≤ 2 Å

Learn structure by reconstructing genes we deliberately hide.

19,000 genes → 512 hidden → 128 latent features

Denosing objective: Vincent et al. · JMLR 2010 / GSE212041 cohort



THE OBJECTIVE

$L = \text{mean } (\hat{x}_i - x_i)^2, i \in \text{masked genes}$

Hide 15% of inputs; score only their reconstruction.

The encoder must use relationships among genes.

149 TRAIN / 74 DEV / 151 EVAL
L40S · 100 epochs · 3.41s fit

TRAIN loss 0.116 vs ≈ 1 for the mean predictor. Generalization is tested on the next slide.

Evidence must grow on data the model has never seen.

Train on the past → bet on a fresh pair → accumulate evidence

Pandeva et al. · AISTATS 2024 / DAVT adaptation

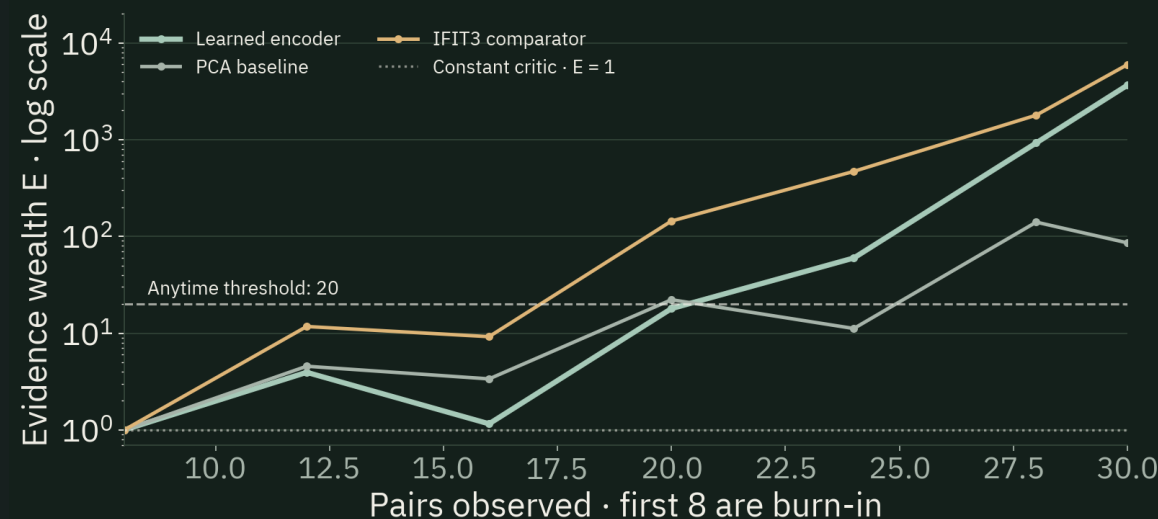
$$L_{\text{bet}} = -\text{mean log } q \quad q = 1 + \tanh(d)$$

$$d = \text{clip}(g(x) - g(y), -4, 4)$$

$$E_t = E_{t-1} \times q_t$$

Under the registered null, swapping x and y makes the bet fair. Arbitrary scores cannot claim that guarantee.

FINAL E: LEARNED 3,653 / PCA 86 / IFIT3 5,964

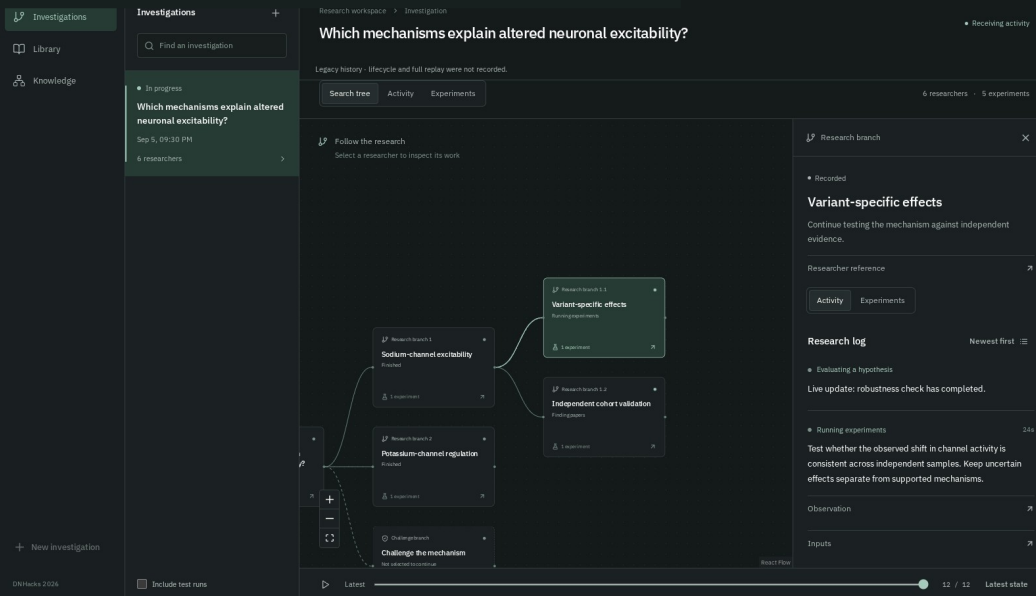


112 / 10,000 null runs ever crossed $E \geq 20$ · 5% anytime error budget under stated assumptions

Trace an investigation, then reveal the later discovery.

Source claim → untested bridge → executable analysis → result → checks → human review

WEBSITE PLACEHOLDER · SAMPLE RUN



[Proposed → passed → reviewed] · [candidate + result]

RESEARCH ARTICLE / LATER PUBLICATION

[Paper title matching
this recovered finding]

[Publication date · DOI]

MATCHED FINDING

[The specific result shared with the candidate]

Backtest discovery against a recorded evidence cutoff.

CAPTURE PLACEHOLDER

Library + source reader

[Capture the actual accessible corpus]
[Open the source claim and quotation]
[Keep the excluded paper outside this view]

FREEZE BEFORE EXECUTION

[Run ID · cutoff · corpus / data hashes]

EXCLUDE AND AUDIT

Target paper withheld; retain retrieval and tool logs

RECORD COMPUTATION

[GPU task · device · time] + saved code / outputs

Next proof: sealed post-cutoff data and equal-budget comparisons against retrieval-only and unguided agents.

Deploy first inside a computational research team.

USER

A research lead deciding what to test with limited compute and reviewer time.

First integration

Approved corpus, data permissions, owned tool execution

Unit economics

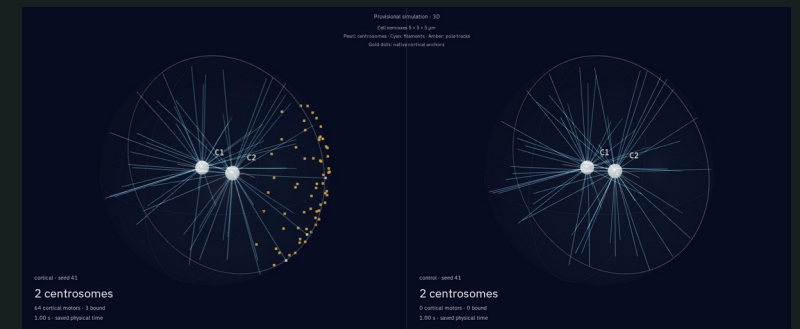
All model + tool + review cost per rerunnable useful result

Adoption test

Review time, repeat use and paid expansion within the team



Tissue: PhysiCell / BioFVM conditional simulation



Spindle: Cytosim mechanics / corrected motor export

Stronger models should produce more science per research budget.

YEAR 1 / 2027

A repeatable workflow

Connect one team's sources and tools.
Establish equal-budget baselines.

Prove: rerunnable findings, review time, cost and repeat use.

YEAR 2 / 2028

Allocation that learns

Train on completed trajectories.
Calibrate on independent episodes.

Prove: discovery yield under a fixed budget; paid expansion.

YEAR 3 / 2029

Continuous programs

Coordinate supervised portfolios as models and instruments improve.

Prove: retained customers, useful output and scalable margins.